

Diabetes Mellitus and Other Endocrine Diseases

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Diabetes Mellitus

Epidemiology

Diabetes mellitus (DM) is a major cause of morbidity and mortality in the United States. More than 19 million Americans have the disease, and approximately 11 percent of all health care expenditures in the United States were directly attributable to the medical care of diabetes in 2002. Men and women diagnosed with diabetes at age 40 years are expected to lose 12 and 14 life-years, respectively. Major studies show that tight glycemic control decreases retinopathy, neuropathy, nephropathy, and possibly coronary artery disease—all of which remain the major contributors to morbidity and mortality. Likewise, tight glycemic control may have a beneficial effect on a subset of skin-related, diabetes-associated disorders.

Diabetes is characterized by a state of relative or complete insulin deficiency, leading to gross defects in glucose, fat, and protein metabolism. In type 1 diabetes (formerly insulin-dependent DM), insulin insufficiency occurs through a gradual, immune-mediated destruction of β islet cells in the pancreas, marked by autoantibodies. In type 2 diabetes (formerly non-insulin-dependent DM), chronic hyperglycemia results primarily from end-organ insulin resistance, followed by a progressive decrease in pancreatic insulin secretion associated with aging. A fasting blood glucose level of ≥ 126 $\mathrm{mg/dL}$ or a random value of ≥ 200 $\mathrm{mg/dL}$ on two separate occasions confirms the diagnosis of diabetes. Type 2 diabetes has a strong genetic

predisposition and is closely associated with obesity. In both types, abnormalities in insulin and elevated blood glucose levels lead to metabolic, vascular, neuropathic, and immunologic disturbances. Affected organs include the cardiovascular, renal, and nervous systems, the eyes, and the skin.

At a Glance

- The incidence of diabetes in America is increasing steadily with the obesity epidemic.
- Eleven percent of health care expenditures in America are diabetes-related.
- Metabolic abnormalities in glucose and insulin relate directly to diabetic thick skin, limited joint mobility, eruptive xanthomas, and acanthosis nigricans.
- Neuropathy, vasculopathy, and immune dysfunction associated with diabetes contribute directly to lower extremity ulcers and certain cutaneous infections.
- Diabetes-associated skin conditions without a known pathogenesis include: **necrobiosis lipoidica, granuloma annulare, diabetic dermopathy, acquired perforating dermatosis, and bullosis diabeticorum.**

Etiology and Pathogenesis

Nearly all patients with diabetes have cutaneous findings related to their condition. Some diabetes-associated skin conditions result directly from metabolic changes such as hyperglycemia and hyperlipidemia. Progressive damage to the vascular, neurologic, and immune systems also contributes significantly to skin manifestations. The mechanisms underlying other diabetes-related skin conditions remain unknown.

Hyperglycemia leads to nonenzymatic glycosylation (NEG) of various structural and regulatory proteins, including collagen. Although NEG occurs normally with aging, it is markedly accelerated in diabetes. NEG results in the formation of advanced glycation end products (AGEs), which reduce the acid solubility and enzymatic digestibility of cutaneous collagen.